INCH-POUND

MIL-H-13719D
7 February 1992
SUPERSEDING
MIL-H-13719C
17 March 1969

MILITARY SPECIFICATION

HOSE ASSEMBLY, RUBBER, HYDRAULIC BRAKE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two types of hose assemblies for use as flexible connections on hydraulic brake equipment for military vehicles.

1.2 Classification. Hose assemblies shall be of the following types, as specified (see 6.2):

Type I	- Regular expansion (see 6.1).
Type II	- Low expansion (see 6.1 and 6.3.1).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Tank-Automotive Command, ATTN: AMSTA-GDS, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 4720

AMSC N/A
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:12Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-H-775	- Hose, Hose Assemblies; Rubber, Plastic, Fabric, or Metal (Including Tubing) and Associated Hardware: Packaging of.
MIL-F-13927	- Fungus Resistance Test; Automotive Components.
MIL-B-46176	- Brake Fluid, Silicone, Automotive, All Weather, Operational and Preservative, Metric.

STANDARDS

FEDERAL

FED-STD-H28/2	- Screw Threads Standards for Federal Services.
FED-STD-151	- Metals: Test Methods.
FED-STD-162	- Hose, Rubber, Visual Inspection Guide for.
FED-STD-601	- Rubber: Sampling and Testing.

MILITARY

MIL-STD-190	- Identification Marking of Rubber Products.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Navy Publication and Printing Services Office, Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B633	- Standard Specifications for Electrodeposited Coatings of Zinc on Iron and Steel.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:12Z
Check the source to verify that this is the current version before use.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1137.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE J526 - Brazed Double Wall Low Carbon Steel Tubing.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096-0001.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Hose assemblies furnished under this specification shall be products which are authorized by the qualifying activity for listing in the applicable qualified products list at the time of award of contract (see 4.4 and 6.4).

3.2 Materials. Unless otherwise specified herein, materials shall conform to the applicable drawing or standard (see 4.6.1 and 6.2).

3.2.1 Recycled, virgin and reclaimed materials. There are no requirements for the exclusive use of virgin materials. The use of recycled or reclaimed (recovered) materials is acceptable provided that all other requirements of this specification are met (see 4.6.1 and 6.5.2).

3.2.2 Rubber. Rubber shall be a nonblooming synthetic compound as specified (see 4.6.1).

3.2.3 Tube. Tube shall be seamless and made from compounds utilizing a copolymer product of butadiene and styrene or polymerized chloroprene as the basic material (see 4.6.1).

3.2.4 Cover. Cover shall be black, made of compounds utilizing polymerized chloroprene as the basic material, and shall be free of holes exposing braid (see 4.6.1 and 4.6.2).

3.3 Design and construction. The design and construction of the hose assemblies shall conform to the applicable drawing or standard (see 4.6.1 and 6.2).

3.3.1 Reinforcement. Reinforcement shall consist of two braids of material imbedded in and bonded to the rubber and shall meet the performance requirements of this specification (see 4.6.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:12Z
Check the source to verify that this is the current version before use.

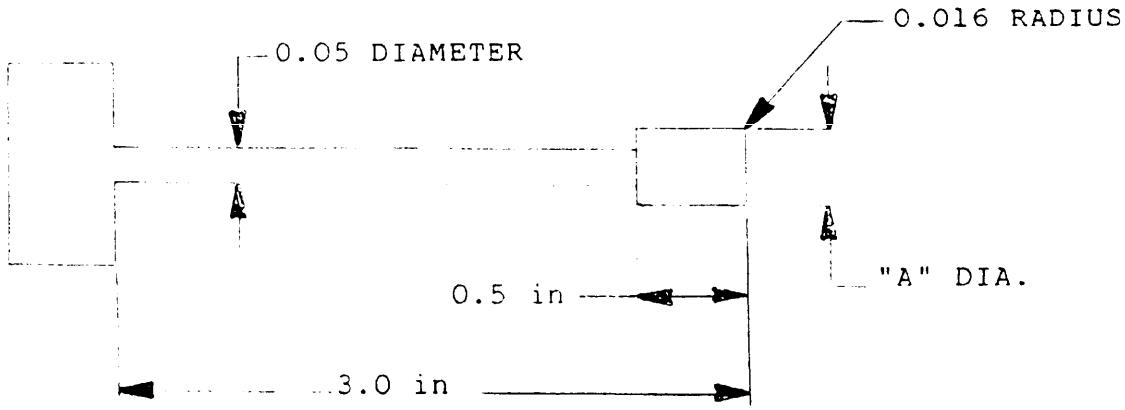
3.3.2 Fittings. Fittings of hose assemblies shall be securely attached to the hose. Dimensions and materials of fittings shall conform to the applicable drawing or standard. Steel fittings shall be zinc coated to conform to class Fe/Zn 12, of ASTM B633. Threads on fittings shall conform to FED-STD-H28/2. Form, threads per inch, and class of fit shall be as specified in applicable drawing or standard (see 4.6.1 and 4.6.2).

3.3.3 Dimensions. Inside diameter (nominal) of the hose shall be 1/8 inch, 3/16 inch, or 1/4 inch. Hose assemblies shall be furnished in the nominal inside diameter and length as specified (see 4.6.2 and 6.2).

3.4 Performance.

3.4.1 Pressure resistance. Hose assemblies shall show no evidence of leakage when subjected to an internal pressure of 1500 psi (minimum) for air or 3000 psi (minimum) for liquid (see 4.6.3).

3.4.2 Constriction. After assembly, the bore of the hose assemblies shall not be constricted. An unconstricted bore shall allow a plug gage conforming to figure 1 to drop through the hose assembly bore of its own weight in less than 5 seconds (see 4.6.4).



HOSE ID	1/8 inch	3/16 inch	1/4 inch
"A" DIAMETER ± 0.002	0.080	0.120	0.165
GAGE WEIGHT	2.020 ± 0.102 OUNCE		

FIGURE 1. Plug gage.

3.4.3 Expansion resistance. Hose shall expand not more than the values shown in table I (see 4.6.5).

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:12Z
Check the source to verify that this is the current version before use.

TABLE I. Maximum hose expansion.
cubic centimeters (c.c.) per foot free length

Hose size Inch I.D.	1000 PSI		1500 PSI	
	Reg. expansion hose type I	Low expansion hose type II	Reg. expansion hose type I	Low expansion hose type II
1/8	0.66	0.33	0.79	0.42
3/16	0.88	0.55	1.02	0.72
1/4	1.04	(a)	1.30	(a)

(a) At present there is no 1/4 inch I.D. low expansion hose available.

3.4.4 Bursting strength. Hose assemblies shall withstand an internal hydrostatic pressure for the 2-minute hold period specified in table II without bursting and shall not leak or burst at any pressure up to and including the burst value specified in table II (see 4.6.6).

TABLE II. Hold and burst pressures.

Nominal ID of hose	Hydrostatic pressure	
	2-minute hold	Minimum burst
Inch	PSI	PSI
1/8	4,000	5,000
3/16	3,000	4,500
1/4	3,000	4,500

3.4.5 Fatigue life. Hose assemblies shall have a minimum life of 35 hours (see 4.6.7).

3.4.6 Low temperature flexibility. Hose assemblies shall show no evidence of breaks, cracks, or leakage (see 4.6.8).

3.4.7 Tensile strength. Hose assemblies shall evidence no rupture of the hose or loosening of the end fittings (see 4.6.9).

3.4.8 Ozone resistance. Hose shall show no evidence of cracks or breaks (see 4.6.10).

3.4.9 Fluid resistance. The volume of the tube specimen shall not increase by more than 20 percent nor decrease by more than 3 percent (see 4.6.11).

3.4.10 Fungus resistance. Hose assemblies shall meet the requirements of 3.4.1 (see 4.6.12).

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:12Z
Check the source to verify that this is the current version before use.

3.4.11 Corrosion resistance. Fittings shall show not more than one red rust spot 1/16 inch in diameter. No white corrosion shall occur on areas that can be contacted by a 0.5 inch diameter ball (see 4.6.13).

3.5 Marking.

3.5.1 Hose assemblies. Hose assemblies, having a free length of 12 inches or longer, shall have the outer cover marked with an oil-resistant, continuous blue stripe. The following information shall be legibly marked or broken into the stripe at repeated intervals of not more than 12 inches on the lay line of the hose (see 4.6.2):

Specification number.
Size (nominal diameter).
Date of manufacture (quarter and year).
Capital letters "OZ".
Manufacturer's code symbol.

3.5.2 Short lengths. Hose assemblies having a free length of less than 12 inches shall have the following information stamped on the fittings (see 4.6.2):

Manufacturer's code.
Capital letters "OZ".

The unit package containing the hose shall, in addition to other required marking, show the following:

Specification number.
Date of manufacture (quarter and year).
Size (nominal diameter).

3.5.3 Rough hose cover. Hose assemblies having a free length of 12 inches or larger, meeting all other requirements of this specification, may, at the discretion of the Government inspector, be marked as specified in 3.5.2, when the cover is too rough, or for any other reason, it is determined that the use of a stripe is impractical on the outside cover. Permanence and legibility of markings shall conform to MIL-STD-190.

3.6 Condition. All hose assemblies offered to the Government for acceptance shall be in good condition and free of defects and shall not have been used or subjected to any tests except as specified in 4.6.3. Hose inner tubing used in the hose assemblies shall have been manufactured not more than 2 quarters prior to presentation for acceptance (see 4.6.1 and 4.6.2).

3.7 Workmanship. Workmanship shall be such as to produce hose assemblies that are free of imperfections, such as, cuts, breaks, blisters, and pitting. Swivel type fittings shall move freely. Seating surfaces of all fittings shall be free of damage (see 4.6.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:12Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examination and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance shall comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations is an acceptable practice to ascertain conformance to requirements, however this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspection: The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.4).
b. Quality conformance inspection (QCI) (see 4.5).
    1. Examination (see 4.5.2).
    2. Tests (see 4.5.3).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed under the following conditions:

a. Air temperature 77 ± 15°F.
b. Barometric pressure 28.5 (+2, -3) inches of mercury.
c. Relative humidity 50 ± 30%.

4.4 Qualification inspection. A qualification sample of sixteen (16) hose assemblies, 1/8 inch I.D. (see 6.3), and a free hose length of 14 inches (measured between fittings) shall be furnished for qualification inspection. Hose assemblies shall be representative of units proposed to be furnished under contract. Qualification inspection shall be conducted under Government surveillance by the contractor, or by an authorized testing facility at a site approved by the Government. Qualification inspection shall consist of all the examinations and tests of this specification, as specified in table III.

4.4.1 Qualification test sequence. Qualification tests shall be conducted on the sample in accordance with the test sequence specified in table IV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:12Z
Check the source to verify that this is the current version before use.

TABLE III. Classification of inspections.

Title	Requirement	Inspection	Qualifications	QCI	
				Examination	Acceptance tests
Group A:					
Materials, design and construction	3.2 thru 3.2.4, 3.3.2 and 3.6	4.6.1	X		
Defects	3.3.1 thru 3.3.3, 3.5 and 3.7	4.6.2	X	X	
Group B:					
Pressure resistance	3.4.1	4.6.3	X		
Constriction	3.4.2	4.6.4	X		X
Expansion resistance	3.4.3	4.6.5	X		
Bursting strength	3.4.4	4.6.6	X		X
Fatigue life	3.4.5	4.6.7	X		
Low temperature flexibility	3.4.6	4.6.8	X		
Tensile strength	3.4.7	4.6.9	X		
Ozone resistance	3.4.8	4.6.10	X		
Fluid resistance	3.4.9	4.6.11	X		
Fungus resistance	3.4.10	4.6.12	X		
Corrosion resistance	3.4.11	4.6.13	X		

TABLE IV. Qualification test sequence.

Sample No.	Paragraph No.
1, 2, 3, 4	4.6.3, 4.6.4, 4.6.7
5, 6	4.6.5, 4.6.6, 4.6.11
7, 8	4.6.10
9, 10	4.6.8, 4.6.4
11, 12	4.6.9
13, 14, 15, 16	4.6.12, 4.6.4, 4.4.13

4.5 QCI. QCI shall include the examination of 4.5.2 and the tests of 4.5.3. Noncompliance with any of the specified requirements in section 3 and 5 shall be cause for rejection of the sample and the inspection lot.

4.5.1 Sampling plan. Unless otherwise specified (see 6.2), the sampling plan specified herein shall be used. See 6.4 for definitions of sampling inspection terms.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:12Z
Check the source to verify that this is the current version before use.

4.5.1.1 Lot formation. An inspection lot shall consist of all units of a one type and part identification number (PIN), from an identifiable production period, from one manufacturer, submitted at the same time for acceptance.

4.5.1.2 Sample. The sample for QCI examination and tests shall be randomly selected from the inspection lot in accordance with table V.

TABLE V. Sampling plan for QCI.

QCI sampling plan			
Inspection lot size	Examination		Test
	Major	Minor	
2 to 8	*	5	2
9 to 15	13	5	2
16 to 25	13	5	3
26 to 50	13	5	5
51 to 90	13	7	5
91 to 150	13	11	8
151 to 280	20	13	13
281 to 500	29	16	13
501 to 1200	34	19	20
1201 to 3200	42	23	32
3,201 to 10,000	50	29	32
10,001 to 35,000	60	35	50
35,001 to 150,000	74	40	80
150,001 to 500,000	90	40	80
500,001 and over	102	40	125

* Indicates entire lot must be inspected (100% inspection).

4.5.2 Examination. The sample selected in accordance with 4.5.1.2 shall be examined and defects classified as specified in table VI. The acceptance number in all cases is zero.

TABLE VI. Classification of defects.

Category	Defect	Method of examination
Critical	None	
Major		
101	Holes in cover exposing braid (see 3.2.4).	Visual
102	Reinforcement braids not imbedded in and bonded to the rubber (see 3.3.1).	Visual
103	Steel fittings not properly coated (see 3.3.2).	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:12Z
Check the source to verify that this is the current version before use.

TABLE VI. Classification of defects - Continued.

Category	Defect	Method of examination
104	Dimensional nonconformance of fittings affecting the interchangeability (see 3.3.2).	SIE 1/
105	Fittings not securely attached to hose (see 3.3.2).	Visual
106	Assembly length incorrect (see 3.3.3).	SIE
107	Diameter nonconformance (see 3.3.3).	SIE
108	Tubing overage (see 3.6).	Visual
109	Faulty workmanship affecting performance (see 3.7).	Visual
Minor		
201	Improper marking (see 3.5).	Visual
202	Illegible marking (see 3.5).	Visual
203	Faulty workmanship affecting appearance (see 3.7).	Visual

1/ SIE = Standard Inspection Equipment.

4.5.3 Acceptance test and sequence. The sample selected in accordance with 4.5.1.2 shall be subjected to the tests specified in table III and in the test sequence specified in table VII. The acceptance number in all cases is zero.

TABLE VII. Acceptance test sequence.

Type	Test sequence
Constriction	4.6.4
Burst	4.6.6

4.5.4 QCI failure. Any item that fails to conform to any specified requirement shall be rejected; any failure (one or more) of the selected sample in either the Major/Minor categories or test for the appropriate inspection lot size shall constitute a failure of the entire lot. The rejected item(s) may be repaired or corrected and resubmitted for inspection. If the contractor utilizes sampling inspection as an element of his inspection system, rejected inspection lots may be resubmitted for acceptance if the contractor performs 100 percent inspection on the lot for those characteristics which were defective and resulted in rejection of the lot and removes all defective units or obtains procuring activity approval to resample the lot due to the insignificance of the defects. Resubmitted lots shall be kept separate from new lots and shall be clearly identified as resubmitted lots.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:12Z
Check the source to verify that this is the current version before use.

4.6 Methods of inspection.

4.6.1 Materials, design and construction. Conformance to 3.2 thru 3.2.4, 3.3.2, and 3.6 shall be determined by inspection of contractor records providing proof or certification that design, construction, processing, and materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

4.6.2 Defects. Conformance to 3.2.4, 3.3.1 thru 3.3.3, 3.5, 3.6 and 3.7, shall be determined by examination for defects listed in table VI. Examination shall be visual, tactile, or by measurement with standard inspection equipment. Visual examination shall be in accordance with FED-STD-162.

4.6.3 Pressure test resistance. To determine conformance to 3.4.1, hose assemblies shall be tested using air or water as the pressure medium. Test pressure shall be 1500 pounds per square inch (psi) minimum for air, and 3000 psi minimum for liquid. CAUTION. Special care should be taken when air is used, as under the pressure specified, air is explosive if a failure should occur in the hose or hose assembly. The pressure shall be applied for not less than 10 nor more than 25 seconds.

4.6.4 Constriction test. To determine conformance to 3.4.2, hose assemblies shall be tested for constriction of the bore by means of the plug gage shown in figure 1. The hose assembly shall be held in a vertical position and the "A" diameter end of the gage inserted into the end of the fitting. The gage shall be permitted to drop through its own weight into the hose assembly.

4.6.5 Expansion test.

4.6.5.1 Apparatus. Test apparatus shall include the following: a source for the required fluid pressures, pressure gages, piping, valves, fittings in which the hose assembly can be mounted in a vertical position for application of pressure under controlled conditions, and a graduated burette for measuring the volume of liquid corresponding to the expansion of the hose under the pressure. All piping and connections shall have a bore devoid of roughness and without recesses or offsets, so that all air can be freely removed from the system before running each test. The valves shall be of such design as to open and close with minimum displacement of liquid. The apparatus shall be capable of applying the pressure at a rate of increase of 25,000 ± 10,000 psi per minute.

4.6.5.2 Calibration. The apparatus shall be tested prior to use to determine its calibration correction factor. For this purpose, 1/4-inch diameter steel tubing conforming to SAE J527 shall be used in place of the hose assembly. Calibration correction factors shall be established at pressures of 1,000 and 1,500 psi, and these shall be subtracted from the expansion readings obtained on the test samples. The maximum permissible calibration correction factor shall be 0.08 cc at 1,500 psi.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:12Z
Check the source to verify that this is the current version before use.

4.6.5.3 Procedure. To determine conformance to 3.4.3, the hose assembly shall be carefully threaded into position on the apparatus in such a way as to provide a leakproof seal, care being taken to avoid twisting, and to maintain the hose in a vertical position without tension while under pressure. Water shall be run through the hose until all air bubbles are removed. The pressure in the hose shall be raised to 1,500 psi for not more than 10 seconds and then released. The water level in the burette shall then be set at zero; and the pressure in the hose shall be raised to 1,000 psi sealed in the hose, and then released by opening the upper valve allowing the water in the expanded hose to rise in the burette. As soon as the liquid level is constant, the upper valve shall be closed and the reading taken on the burette. This operation shall be repeated so that the final reading taken on the burette shall be the total of three expansions. This final reading divided by three, minus the calibration factor, shall be considered as the final expansion of the hose at 1,000 psi. The water level in the burette shall be readjusted to zero as above and the procedure repeated at 1,500 psi to obtain the expansion at that pressure.

4.6.6 Bursting test. To determine conformance to 3.4.4, hose assemblies shall be tested as specified in method 10011 of FED-STD-601, except the pressure shall be raised to the hold pressure specified in table II. The pressure shall then be increased until the minimum burst pressure, as specified in table II, has been reached.

4.6.7 Fatigue life (whip) test.

4.6.7.1 Apparatus. Test machine shall include two parallel, horizontal headers, each provided with four standard end connections spaced 3-1/2 inches apart to mount four hose assemblies in parallel. One header shall be ball bearing mounted at the ends on disks, which rotate in the same vertical plane, so that the attached hose assembly ends describe 8-inch diameter circles. The other header shall be horizontally adjustable in the plane of the disk centers to accommodate hose assemblies of different lengths. The end connections to the moving header shall effectively cap the hose assemblies, while those on the stationary header shall be open to a manifold, through which water pressure shall be applied by means of a weight operated plunger in a pressure cylinder. A limit switch operated by the plunger weight shall stop the machine when hose failure causes the water pressure to fall. A revolution counter and time interval recorder shall be provided or a machine that provides similar motion.

4.6.7.2 Procedure. To determine conformance to 3.4.5, hose assemblies shall be mounted in the test machine so that the horizontal projected length is less than their straight length by the amount indicated as slack in table VIII. Hose assemblies shall be installed in the machine in their natural lay, that is, without any twist. When all four hose assemblies are installed, water pressure shall be applied and all hoses and passages bled to eliminate air pockets and bubbles. During the test, a constant pressure between 225 and 250 psi shall be maintained. The disks shall revolve at 800 ± 10 revolutions per minute (rpm). Failure of any hose assembly by water leakage through rupture, and consequent loss of pressure, shall terminate the test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:12Z
Check the source to verify that this is the current version before use.

TABLE VIII. Slack required for fatigue life (whip) test.

Free length between fittings	Slack required		
	1/8 inch diameter	3/16 inch diameter	1/4 inch diameter
Inches 14	Inches 1.750 ± 0.015	Inches 1.000 ± 0.015	Inches 1.000 ± 0.015

4.6.8 Low temperature flexibility test. To determine conformance to 3.4.6, hose assemblies shall be conditioned for 72 hours at a temperature of minus 85 ± 4°F. At the end of this period and while still at the specified temperature, the sample shall be bent, within 10 seconds, through 180° from the centerline over a mandrel with a diameter equal to 10 times the nominal outside diameter (OD) of the hose. While still at the specified temperature, the sample shall then be bent through 360° in the opposite direction over the same mandrel within the next 10 seconds. After completion of the bending at the specified temperature, the sample shall be subjected to the test specified in 4.6.3.

4.6.9 Tensile strength test. To determine conformance to 3.4.7, the hose assembly shall be attached to a testing machine by means of the two hose assembly fittings, and subjected to a straight pull at a speed of 1 inch per minute, until a pull of 325 pounds is attained.

4.6.10 Ozone resistance test.

4.6.10.1 Testing apparatus. Apparatus for performing the test shall include an exposure chamber that will maintain a mixture of air and ozone in the proportion of 50 ± 5 parts of ozone per 100 million parts of air by volume. The chamber shall be capable of maintaining an air temperature of 100 ± 2°F during the test.

4.6.10.2 Procedure. To determine conformance to 3.4.8, the test shall be conducted in an ambient temperature of 100 ± 5°F. The sample shall be bent around a mandrel, the diameter of which shall be 7 times the OD of the hose being tested, and bound with twine or tape at the point where the ends of the hose cross one another. If collapse of the hose occurs, provision shall be made to support the hose internally. The sample shall be conditioned 45 minutes in air at room temperature. The sample, while still on the mandrel, shall be placed into the chamber and exposed to the mixture of ozone and air for a period of 168 hours. At the end of the 168-hour period, the hose cover shall be examined under 9-power magnification, ignoring the areas immediately adjacent to or within the area of the sample where taped or tied.

4.6.11 Fluid resistance test. To determine conformance to 3.4.9, sample of tube shall be tested as specified in method 6211 of FED-STD-601, except that the aging shall be for 70 hours at a temperature of 212 ± 5°F. The immersion fluid shall conform to MIL-B-46176.

Source: https://assist.dla.mil -- Downloaded: 20261O-18T02:12Z
Check the source to verify that this is the current version before use.

4.6.12 Fungus resistance test. To determine conformance to 3.4.10, hose assemblies shall be tested in accordance with class 3, method A of MIL-F-13927 then tested as specified in 4.6.3.

4.6.13 Corrosion resistance test. To determine conformance to 3.4.11, hose assemblies shall be subjected to a 96-hour salt spray in accordance with method 811.1 of FED-STD-151.

5. PACKAGING

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking shall be in accordance with MIL-H-775 (see 6.2).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The hose assemblies furnished under this specification are for use as flexible connections on automotive hydraulic brake components. Type I hose may be used in applications requiring regular expansion hose. Type II hose may be used in applications requiring either low expansion or regular expansion hose.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type hose required (see 1.2).
c. Applicable drawings (see 3.2), and length and ID of hose assemblies (see 3.3.3).
d. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
e. If responsibility for inspection shall be other than as specified (see 4.1).
f. If inspection conditions shall be other than as specified (see 4.3).
g. If sampling plan for QCI is other than as specified (see 4.5.1).
h. Selection of applicable level and packaging requirements (see 5.1).

6.3 Special provisions for establishing qualification. Hose assemblies submitted by a manufacturer should be 1/8 inch. Upon being qualified, the 1/8 inch-size hose assembly should then be recognized as establishing qualification of all other hose assembly sizes by that same manufacturer.

6.3.1 New type II hose. Successful completion of testing of type II hose will establish qualification of type I hose. If the situation in the industry develops as expected, it is probable that type I hose will be completely replaced by type II hose at some future date. If this happens, all reference to types may be removed from the specification; in which case any drawing referencing type I or type II will automatically refer to the hose described in the current revision of the specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:12Z
Check the source to verify that this is the current version before use.

6.4 Qualification. With respect to products requiring qualification, award will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable QPL, whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the QPL is the U.S. Army Tank-Automotive Command, ATTN: AMSTA-GDS, Warren, Michigan 48397-5000, and information pertaining to qualification of products may be obtained from that activity.

6.5 Definitions.

6.5.1 Definitions of terms used in sampling inspection.

a. Classification of defects. A classification of defects is the enumeration of possible defects of the unit of product classified according to their seriousness. A defect is any nonconformance of the unit of product with specified requirements. Defects will normally be grouped into one or more of the following classes: critical, major and minor defects. Also, defects may be grouped into other classes, or into subclasses within these classes.

b. Critical defects. A critical defect is a defect that judgement and experience indicate would result in hazardous or unsafe conditions for individuals using, maintaining, or depending upon the product, or a defect that judgement and experience indicate is likely to prevent performance of the tactical function of a major end item such as a ship, aircraft, tank, missile, or space vehicle.

c. Critical defective. A critical defective is a unit of product which contains one or more critical defects and may also contain major and/or minor defects.

d. Defective. A defective is a unit of product which contains one or more defects.

e. Formation of lots or batches. The product shall be assembled into identifiable lots, sublots, batches, or in such other manner as may be prescribed (see 1). Each lot or batch shall, as far as is practicable, consist of units of product of a single type, grade, class, size, and composition, manufactured under essentially the same conditions, and at essentially the same time.

f. Lot or batch. The term lot or batch shall mean "inspection lot" or "inspection batch", i.e., a collection of units or product from which a sample is to be drawn and inspected and may differ from a collection of units designated as a lot or batch for other purposes (e.g., production, shipment, etc.).

g. Lot or batch size. The lot or batch size is the number of units of product in a lot or batch.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:12Z
Check the source to verify that this is the current version before use.

h. Major defect. A major defect is a defect, other than critical, that is likely to result in failure, or to reduce materially the usability of the unit of product for its intended purpose.

i. Major defective. A major defective is a unit of product which contains one or more major defects, and may also contain minor defects but contains no critical defect.

j. Minor defect. A minor defect is a defect that is not likely to reduce materially the usability of the unit of product for its intended purpose, or is a departure from established standards having little bearing on the effective use or operation of the unit.

k. Minor defective. A minor defective is a unit of product which contains one or more minor defects but contains no critical or major defect.

l. Presentation of lots or batches. The formation of the lots or batches, lot or batch size, and the manner in which each lot or batch is to be presented and identified by the supplier shall be designated or approved by the responsible authority. As necessary, the supplier shall provide adequate and suitable storage space for each lot or batch, equipment needed for proper identification and presentation, and personnel for all handling of product required for drawing of samples.

m. Representative sampling. When appropriate, the number of units in the sample shall be selected in proportion to the size of sublots or subbatches, or parts of the lot or batch, identified by some rational criterion. When representative sampling is used, the units from each part of the lot or batch shall be selected at random.

n. Sample A sample consists of one or more units of product drawn from a lot or batch, the units of the sample being selected at random without regard to their quality. The number of units or product in the sample is the sample size.

o. Sampling plan. A sampling plan indicates the number of units of product from each lot or batch which are to be inspected (sample size or series of sample sizes) and the criteria for determining the acceptability of the lot or batch (acceptance and rejection numbers).

p. Time of sampling. Samples may be drawn after all the units comprising the lot or batch have been assembled, or samples may be drawn during assembly of the lot or batch.

6.5.2 Recovered materials. "Recovered materials" means materials that have been collected or recovered from solid waste (see 6.5.3).

6.5.3 Solid waste. "Solid waste" means (a) any garbage, refuse, or sludge from a waste treatment plant, water supply treatment plant, or air pollution control facility; and (b) other discarded material, including solid, liquid, semisolid, or contained gaseous material resulting from industrial, commercial, mining, and agricultural operations, and from

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:12Z
Check the source to verify that this is the current version before use.

community activities. It does not include solid or dissolved material in domestic sewage, or solid or dissolved material in irrigation return flows or industrial discharges which are point sources subject to permits under section 402 of the Clean Water Act, (33 U.S.C. 1342 et seq.), or source, special nuclear, or byproduct material as defined by the Atomic Energy Act of 1954 (42 U.S.C. 2011 et seq.) (Source: Federal Acquisition Regulations, section 23.402).

6.6 Part or identifying number (PIN). The PIN to be used for hose assemblies acquired to this specification are created as follows:

M - 13719 - X

- X: Type (see 1.2):
  - 1 - Type I, Regular expansion.
  - 2 - Type II, Low expansion.
- 13719: Specification number.
- M: Prefix to indicate military specification.

6.7 Subject term (key word) listing.

Hose
Hose assembly
Hydraulic brake
Rubber

6.8 AMC policy on AQLs/LTPDs. This specification is certified to be in compliance with current Army Materiel Command (AMC) policy for the elimination of AQLs/LTPDs (Acceptable Quality Levels/Lot Tolerance Percent Defectives) from military specifications.

6.9 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - AT

Review activities:
Army - MI
DLA - CS

User activities:
Army - ME, SM
Navy - MC

Preparing activity:
Army - AT

(Project 4720-0026)

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:12Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.

NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. FSC 4720

DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-H-13719D; Hose Assembly, Rubber, Hydraulic Brake

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

DD FORM 1426 1 OCT 76

EDITION OF 1 JAN 72 WILL BE USED UNTIL EXHAUSTED

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:12Z
Check the source to verify that this is the current version before use.